

Definitions

- **Stationary:** A time series where statistical properties don't change over time
 - E.g. mean and variance and consistent over time

Time2Vec

- **Pros**
 - Flexible -> works well in nonlinear/non-stationary/irregular variability settings
 - Is able to recognize more complicated trends (for example the asymmetry in light curves)
- **Cons**
 - Less Interpretable, (in our case, we can't directly translate the values found to the actual period of the light curve)
 - The model is much more focused on predicting values instead of on calculating period
 - Needs a lot of data to become accurate
 - Not clear what exactly "a lot" is
 - Bad at recognizing noise
 - Susceptible to overfitting
- **Use Cases**
 - When working with complex, non-linear, or non-stationary patterns that are difficult to model manually
 - When dealing with asynchronous, irregularly sampled, or limited historical data
 - When model performance is the primary goal and interpretability is a secondary concern
 - In our case, this means if we don't want to use the model to determine period

Fourier

- **Pros**
 - Interpretable: Every coefficient has a very clear meaning
 - In our case, this means we can clearly correlate coefficients to the periods of variable stars
 - Efficient
 - Resilient to noise (filters it out)
 - Lots of past research on implementing it in our context
- **Cons**
 - Doesn't work for non-stationary data
 - Doesn't work well for irregularly sampled data (although this can be fixed with certain adjustments)
 - Sounds like Lomb-Scargle might have a fix for this
- **Use Cases**
 - When working with stationary data with clear, fixed periodic components
 - From what I'm seeing online it sounds like LPV periods might drift? Need to talk more about this

- When you need to filter out noise

Conclusion

For RR Lyraes it seems like the clear superior choice is Fourier. Every sign points to that from what I can tell. For LPVs, depending on how we feel about LPV period drift, we may want to consider Time2Vec or some sort of combination of Fourier and Time2Vec (I'm not sure exactly how this is implemented, but it sounds like it might be a way to get some of the benefits from each and bridge the gap between the two).